

Strong Perversity of Tautological Classes & Global Springer Theory.

Recollection of previous lectures:

C : irreducible smooth proj. curve, $g \geq 2 / \mathbb{C}$. $(n, d) = 1$

Conj 0 ($P=W$ for GL)

homeo. by Simpson corresp. $M_{n,d}^{\text{Dol}}$ = moduli space of stable Higgs bundles of (n, d)
 $M_{n,d}^B$ = twisted character variety.

$$\text{conjecture says } P_k H^*(M_{n,d}^{\text{Dol}}; \mathbb{Q}) = W_{2k} H^*(M_{n,d}^B; \mathbb{Q}) \\ = W_{2k+1} H^*(M_{n,d}^B; \mathbb{Q})$$

It is equivalent to PGL_n -version:

non-abelian Hodge isom between $H^*(-; \mathbb{Q})$ $\cdot \hat{M}_{n,d}^{\text{Dol}} = \check{M}_{n,d}^{\text{Dol}} / (\text{Pic}^d(C)[n])$ [a DM-stack, but NOT a variety]
 $\check{M}_{n,d}^{\text{Dol}} = \langle (\mathcal{E}, \theta) \mid \det(\mathcal{E}) \cong \mathcal{N} \text{ (fixed in } \text{Pic}^d(C), \text{tr}(\theta) = 0 \rangle$
 (Also refer to SL_n -Dolbeault (Higgs) moduli)
 $\cdot \hat{M}_{n,d}^B$ = twisted PGL_n -character variety.

$$H^*(\hat{M}_{n,d}^{\text{Dol}}; \mathbb{Q}) \longrightarrow H^*(\hat{M}_{n,d}^B; \mathbb{Q}) \quad (*)$$

Conj 1 ($P=W$ for PGL_n)

$$P_k H^*(\hat{M}_{n,d}^{\text{Dol}}; \mathbb{Q}) = W_{2k} H^*(\hat{M}_{n,d}^B; \mathbb{Q}). \\ = W_{2k+1} H^*(\hat{M}_{n,d}^B; \mathbb{Q}) \quad (\text{by the following})$$

$\mathcal{U}$: tautological PGL_n -bundle on $C \times \hat{M}^{\mathrm{Dol}}$

$$\begin{array}{ccc} & \text{on } C \times \hat{M}^{\mathrm{Dol}} & \\ & \swarrow p_1 \quad \searrow p_2 & \\ C & & \hat{M}^{\mathrm{Dol}} \end{array}$$

$$\Rightarrow \mathrm{ch}_k(\mathcal{U}) \in H^{2k}(C \times \hat{M}^{\mathrm{Dol}}, \mathbb{Q}) \quad k \geq 2$$

$$C_k(\gamma) := \int_{\gamma} \mathrm{ch}_k(\mathcal{U}) = p_{2*} (p_1^* \gamma \cup \mathrm{ch}_k(\mathcal{U})) \in H^*(\hat{M}^{\mathrm{Dol}}, \mathbb{Q})$$

$$\forall \gamma \in H^*(C, \mathbb{Q})$$

Thm (Markman + Shende) [Emmanuel + Benjamin's talk]

(i) $C_k(\gamma)$ generates $H^*(\hat{M}^{\mathrm{Dol}}, \mathbb{Q})$.

(ii) Under non-abelian Hodge (*),

$$C_k(\gamma) \in W_{2k} H^*(\hat{M}^{\mathrm{B}}, \mathbb{Q}) \cap F^k H^*(\hat{M}^{\mathrm{B}}, \mathbb{Q}) =: {}^k \mathrm{Hdg}^*(\hat{M}^{\mathrm{B}})$$

$$H^*(\hat{M}^{\mathrm{B}}, \mathbb{Q}) = \bigoplus_{m,k} {}^k \mathrm{Hdg}^m(\hat{M}^{\mathrm{B}})$$

Also, by curious hard Lefschetz (CHL). [Olivier's talk]

$$W_{2k} \subseteq P_k \quad \text{implies} \quad W_{2k} = P_k$$

Then Conj 1 reduces to show.

$$\text{Conj: } \forall k_i, \gamma_i \quad \underbrace{C_{k_1}(\gamma_1) \cup \dots \cup C_{k_n}(\gamma_n)}_{\text{generates } W_{2\sum k_i} H^*(\hat{M}^{\mathrm{B}}, \mathbb{Q})} \in P_{\sum k_i} H^*(\hat{M}^{\mathrm{Dol}}, \mathbb{Q})$$

$\leadsto$ notion of strong perversity

[Junhui's talk]

& techniques of vanishing cycles.

Def: $f: X \rightarrow Y$ smooth, proper; $\gamma \in H^{\ell}(X; \mathbb{Q})$ has strong perversity

$$c \text{ wrt. } f \text{ if } \gamma: {}^p \tau_{\leq i} Rf_* \mathbb{Q} \rightarrow {}^p \tau_{\leq i+c-\ell} Rf_* \mathbb{Q} [\ell]$$

As a consequence: γ has strong perversity c . then

$$\gamma_U -: P_i H^* \longrightarrow P_{i+c} H^{*+l}$$

$\leadsto$ Reduce to show:

proper [Simpson]
+ smooth $\downarrow$

$c_k(\gamma) \in H^{2k}(\hat{M}^{\text{Dol}}, \mathbb{Q})$ has strong perversity k wrt Hitchin fibration

$\leadsto$ Consider the tautological class $ch_k(\mathcal{U})$:

Want: $ch_k(\mathcal{U}) \in H^{2k}(C \times \hat{M}^{\text{Dol}}, \mathbb{Q})$ has strong perversity k
wrt $\text{id} \times \hat{h}$

According to the transitivity of strong perversity along
vanishing cycle, want:

Main thm:

$\exists$ effective line bundle $\mathcal{L}$ st.

$ch_k(\mathcal{U}^{\mathcal{L}}) \in H^{2k}(C \times \hat{M}^{\text{Dol}}, \mathbb{Q})$ has strong perversity k .
wrt. $\text{id} \times \hat{h}^{\mathcal{L}}$

The proof uses global Springer Theory: [Eric's talk]

(Abbreviate $M^{\text{Dol}}, \hat{M}^{\text{Dol}}, \hat{\mathcal{M}}^{\text{Dol}}$ as $M, \hat{M}, \hat{\mathcal{M}}$)

$G := \text{PGL}_n \supseteq B \supseteq T$ $W \simeq \mathfrak{S}_n$ Weyl group

$X^*(T)$: character lattice $\simeq \mathbb{Z}^{n-1}$

Recall $\hat{\mathcal{M}}_{\mathcal{L}}^{\text{par}}$: parabolic G -Higgs bundles. ($\mathcal{L}$ -twisted)

given by tuple: $\left\langle (x, \mathcal{E}, \theta, \mathcal{E}_x^B) \mid \begin{array}{l} x \in C, (\mathcal{E}, \theta) \in \hat{\mathcal{M}}_{\mathcal{L}} \\ \mathcal{E}_x^B : B\text{-reduction of } \mathcal{E}|_x \\ \text{Compatible w/ } \theta \end{array} \right\rangle$

forgetful map completes to a Cartesian diagram:

$$\begin{array}{ccc}
 (x, \varepsilon, \theta, \varepsilon_x^B) \hat{\mathcal{M}}^{\text{par}} & \xrightarrow{\text{ev}^{\text{par}}} & [b/B]_{\mathcal{L}} = \rho_{\mathcal{L}}^{\mathbb{G}_m} [b/B] \\
 \downarrow & \downarrow \pi & \downarrow \pi_G \text{ twisted Grothendieck-Springer} \\
 (x, \varepsilon, \theta) C \times \hat{\mathcal{M}} & \xrightarrow{\text{ev}} & [a/G]_{\mathcal{L}} = \rho_{\mathcal{L}}^{\mathbb{G}_m} [a/G] \text{ resolut.}
 \end{array}$$

Define $\hat{\mathcal{M}}^{\text{par}} := (C \times \hat{\mathcal{M}}) \times_{C \times \hat{\mathcal{M}}} \hat{\mathcal{M}}^{\text{par}}$

$\hat{h}^p := (\text{id} \times \hat{h}) \circ \pi : \hat{\mathcal{M}}^{\text{par}} \longrightarrow C \times \hat{A}$ (parabolic Hitchin map)

Prop: $\hat{\mathcal{M}}^{\text{par}}$ is a smooth Deligne-Mumford stack, and the parabolic Hitchin map $\hat{h}^p$ is proper. (Assume $\deg(\mathcal{L}) \gg 0$?)

$$\begin{array}{ccccccc}
 & & \hat{\mathcal{M}}^{\text{par}} & \xrightarrow{\quad} & \hat{\mathcal{M}}^{\text{par}} & \xrightarrow{\text{ev}^{\text{par}}} & [b/B]_{\mathcal{L}} \\
 & \swarrow \hat{h}^p & \downarrow \pi & & \downarrow \pi & & \downarrow \pi_G \\
 C \times \hat{A} & \xleftarrow{\text{id} \times \hat{h}} & C \times \hat{\mathcal{M}} & \xrightarrow{\quad} & C \times \hat{\mathcal{M}} & \xrightarrow{\text{ev}} & [a/G]_{\mathcal{L}}
 \end{array}$$

Pf: The properness is clear

$\hat{\mathcal{M}}$ is known as DM-stack, and π is the pull-back of π_G , which is Schematic and of finite type, thus $\hat{\mathcal{M}}^{\text{par}}$ is DM.

Ingredients to prove smoothness: $R \in k\text{-Alg}$. $m \in \hat{\mathcal{M}}^{\text{par}}(R)$
 $(x, \varepsilon, \theta, \varepsilon_x^B)$

is given by the diagram:

$$\begin{array}{ccc}
 \text{Spec } R & \xrightarrow{m_x} & [b/B]_{\mathcal{L}} \\
 x \downarrow & & \downarrow \pi \\
 X_R & \xrightarrow{m} & [a/G]_{\mathcal{L}}
 \end{array}$$

Now take $R \in \text{Artin}_k$, $I \subseteq R$ square-zero ideal. $R_0 = R/I$

Suppose $m_0 = (x_0, \varepsilon_0, \theta_0, \varepsilon_{x_0}^B) \in \hat{\mathcal{M}}^{\text{par}}(R_0)$.

want to lift m_0 to $m \in \hat{\mathcal{M}}^{\text{par}}(R)$. need to find (m_x, m)

s.t.

$$\begin{array}{ccc} X_{R_0} & \xrightarrow{m_0} & [\mathcal{Y}/G]_{\mathcal{L}} \\ \downarrow & \nearrow m & \downarrow \\ X_R & \longrightarrow & X \end{array} \quad \begin{array}{ccc} \text{Spec } R_0 & \xrightarrow{m_{x_0}} & [\mathcal{b}/B]_{\mathcal{L}} \\ \downarrow & \nearrow m_x & \downarrow \\ \text{Spec } R & \xrightarrow{x} & X \end{array}$$

(*)

$$\begin{array}{ccccc} & & \text{Spec } R & \xrightarrow{m_x} & [\mathcal{b}/B]_{\mathcal{L}} \\ & & \swarrow & & \uparrow m_{x_0} \\ & & \text{Spec } R_0 & & \downarrow \\ x \downarrow & & & & \\ X_R & \xleftarrow{x_0} & X_{R_0} & \xrightarrow{m_0} & [\mathcal{Y}/G]_{\mathcal{L}} \\ & \nearrow m & & & \downarrow \end{array} \quad \text{commute.}$$

Infinitesimal deformation m of m_0 (resp. m_x of m_{x_0}) is controlled by

$$R\text{Hom}_{X_{R_0}}(m_0^* \mathbb{L}_{[\mathcal{Y}/G]_{\mathcal{L}}/X}, I \otimes \mathcal{O}_X)$$

(resp.) $R\text{Hom}_{R_0}(m_{x_0}^* \mathbb{L}_{[\mathcal{b}/B]_{\mathcal{L}}/X}, I)$

To fit into diagram (*). the lift of $\hat{\mathcal{M}}^{\text{par}}(R_0)$ to $\hat{\mathcal{M}}^{\text{par}}(R)$ at one point $(x_0, \varepsilon_0, \theta_0, \varepsilon_0^B)$ is controlled by $\mathcal{K}_I$, where

$$\begin{aligned} \mathcal{K}_I &\longrightarrow R\text{Hom}_{X_{R_0}}(m_0^* \mathbb{L}_{[\mathcal{Y}/G]_{\mathcal{L}}/X}, I \otimes \mathcal{O}_X) \oplus R\text{Hom}_{R_0}(m_{x_0}^* \mathbb{L}_{[\mathcal{b}/B]_{\mathcal{L}}/X}, I) \\ &\xrightarrow{(i^*, -\pi^*)} R\text{Hom}_{R_0}(i^* m_0^* \mathbb{L}_{[\mathcal{Y}/G]_{\mathcal{L}}/X}, I) \end{aligned}$$

$i: \Gamma_{x_0} \hookrightarrow X_{R_0}$ π^* given by $\pi^* \mathbb{L}_{[\mathcal{Y}/G]_{\mathcal{L}}/X} \longrightarrow \mathbb{L}_{[\mathcal{b}/B]_{\mathcal{L}}/X}$

$$\left(\begin{array}{ccc} m_{x_0}^* \pi^* \mathbb{L}_{[\mathcal{Y}/G]_{\mathcal{L}}/X} & \longrightarrow & m_{x_0}^* \mathbb{L}_{[\mathcal{b}/B]_{\mathcal{L}}/X} \longrightarrow I \\ i^* m_0^* \mathbb{L}_{[\mathcal{Y}/G]_{\mathcal{L}}/X} & \xrightarrow{\quad} & \end{array} \right)$$

$$[Ng\hat{o}] \quad m_0^* \mathbb{L}_{[\mathfrak{y}/G]_{\mathbb{L}/X}} = [Ad(E_0)^\vee \xrightarrow{Ad(\varphi_0)} Ad(E_0)^\vee(\mathbb{L})]$$

$$m_{x_0}^* \mathbb{L}_{[\mathfrak{b}/B]_{\mathbb{L}/X}} = [Ad(E_{x_0}^B)^\vee \xrightarrow{Ad(\varphi_{lx_0})} Ad(E_{x_0}^B)^\vee \otimes i^* \mathbb{L}]$$

2-term complexes in deg $[-1, 0]$

$$\Rightarrow \mathcal{F} := \text{Ker} \left(Ad(E_0) \longrightarrow i_* (i^* Ad(E_0) / Ad(E_{x_0}^B)) \right)$$

$$\mathcal{K}_I \text{ is given by } [\mathcal{F} \otimes_{R_0} I \longrightarrow \mathcal{F}(\mathbb{L}) \otimes_{R_0} I]$$

Now the obstruction lives in $H^1(X_{R_0}, \mathcal{K}_I)$

Reduce to computation in [Maulik - Shen From Pi Prop 4.1] $\square$

We see that parabolic Hitchin fibration $\hat{h}^P: \hat{M}^{\text{par}} \longrightarrow C \times \hat{A}$

has nice properties, return to the proof of Main thm:

Again $\pi: \hat{M}^{\text{par}} \longrightarrow C \times \hat{M}$, then:

(A): the Chern roots of $\pi^* \mathcal{U}$ are given by $c_1(L(\xi))$

$L(\xi)$: tautological bundle on $\hat{M}^{\text{par}}$ asso. to $\xi \in \chi^*(T)$

(B): $\exists$ Zariski dense open in $C \times \hat{A}$, on which

$$c_1(L(\xi)): R\hat{h}_*^P \mathcal{Q}_{\hat{M}^{\text{par}}} \longrightarrow R\hat{h}_*^P \mathcal{Q}_{\hat{M}^{\text{par}}}$$

has strong perversity 1

(C) Pull-back Weyl group action on Grothendieck resolution.

to get (canonical) W -action on $R\pi_* \mathcal{Q}_{\hat{M}^{\text{par}}}$ s.t.

$$(R\pi_* \mathcal{Q}_{\hat{M}^{\text{par}}})^W = \mathcal{Q}_{C \times \hat{M}}$$

$ch_k(\pi^* \mathcal{U})$
has
strong
perversity k !

[Need a
support thm]

With help of (A), (B), (C), we have

Lem.: $\forall \gamma \in H^l(C \times \hat{M}, \mathbb{Q})$, if $\pi^* \gamma \in H^l(\hat{M}^{\text{par}}, \mathbb{Q})$ has strong perversity k wrt. $\hat{h}^p$, then γ has strong perversity k wrt. $h = \text{id} \times \hat{h}: C \times \hat{M} \longrightarrow C \times \hat{A}$

Pf: $\gamma: \mathbb{Q}_{C \times \hat{M}} \longrightarrow \mathbb{Q}_{C \times \hat{M}}[l]$ pull-back the complex by π^*

$\pi^* \gamma: \mathbb{Q}_{\hat{M}^{\text{par}}} \longrightarrow \mathbb{Q}_{\hat{M}^{\text{par}}}[l]$ recovers $\pi^* \gamma \in H^l(\hat{M}^{\text{par}}, \mathbb{Q})$.

$\pi_* \pi^* \mathbb{Q} = |W| \cdot \mathbb{Q}$ as π generic Galois covering

By projection formula $\pi_* \pi^* \gamma = |W| \cdot \gamma$

$\hookrightarrow \gamma$ is recovered from $\pi^* \gamma$ by pushforward then take the W -invariant part.

Denote also $\pi^* \gamma: R\hat{h}_*^p \mathbb{Q}_{\hat{M}^{\text{par}}} \longrightarrow R\hat{h}_*^p \mathbb{Q}_{\hat{M}^{\text{par}}}[l]$

$Rh_*(R\pi_* \mathbb{Q}_{\hat{M}^{\text{par}}}) \hookrightarrow W\text{-action}$

By (C), the W -invariant part is $\mathbb{Q}_{C \times \hat{M}}$

$\Rightarrow$ recover $\gamma: Rh_* \mathbb{Q}_{C \times \hat{M}} \longrightarrow Rh_* \mathbb{Q}_{C \times \hat{M}}[l]$, which

has strong perversity k wrt. h

$\triangle$

Properties of $\pi: \hat{M}^{\text{par}} \longrightarrow \mathbb{C} \times \hat{M}$

(A) Consider $\text{Bun}_G^{\text{par}} = \langle (x, \mathcal{E}, \mathcal{E}_x^B) \rangle \longleftarrow \hat{M}^{\text{par}}$

There is a tautological T -torsor on $\text{Bun}_G^{\text{par}}$ over $(x, \mathcal{E}, \mathcal{E}_x^B)$
its fiber is given by $\mathcal{E}_x^B/N$. denote by L .

$(\xi: T \longrightarrow G_m) \in X^*(T)$. can be associated w/ a line bundle $L(\xi) := L \times^{T, \xi} G_m$ on $\text{Bun}_G^{\text{par}}$.

pull-back to $\hat{M}^{\text{par}}$. it is a line bundle on $\hat{M}^{\text{par}}$

(C) Consider

$$\begin{array}{ccc} \hat{M}^{\text{par}} & \xrightarrow{\text{ev}^{\text{par}}} & [b/B]_{\mathbb{Q}} \\ \pi \downarrow & & \downarrow \pi_G \\ \mathbb{C} \times \hat{M} & \xrightarrow{\text{ev}} & [g/G]_{\mathbb{Q}} \end{array}$$

Then the W -action on $R\pi_* \mathcal{Q}_{\hat{M}^{\text{par}}}$ is the pull-back of W -action on simultaneous Grothendieck resolution π_G .

($\hat{M}$ versus $\hat{M}^{\text{par}}|_{\hat{A}^\vee} ??$)

and the invariant part $(R\pi_* \mathcal{Q}_{\hat{M}^{\text{par}}})^W = \mathcal{Q}_{\mathbb{C} \times \hat{M}}$

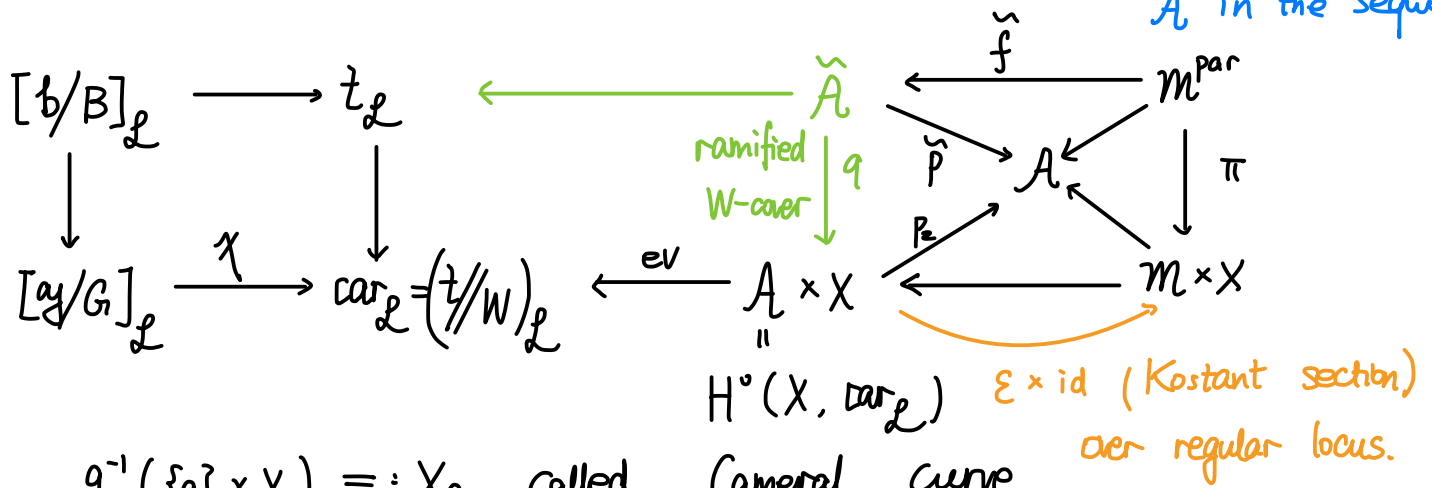
(B) is difficult!

Now $c_1(L(\xi)) \in H^2(\hat{M}^{\text{par}}, \mathbb{Q})$ has naturally strong perversity 2 wrt. $\hat{h}^p$. to show it has strong perversity 1. (over some open dense)

ie $c_1(L(\xi)): {}^p\mathcal{L}_{\xi i} R\hat{h}^p \mathcal{Q}_{\hat{M}^{\text{par}}} \longrightarrow {}^p\mathcal{L}_{\xi i-1} R\hat{h}^p \mathcal{Q}_{\hat{M}^{\text{par}}}[2]$

equivalent to show ${}^p\mathcal{H}^i(R\hat{h}^p \mathcal{Q}_{\hat{M}^{\text{par}}}) \longrightarrow {}^p\mathcal{H}^i(R\hat{h}^p \mathcal{Q}_{\hat{M}^{\text{par}}}[2])$ vanishes $\forall i$

The base is taken to be $\hat{A}^\diamond$. as follows: (we don't specify $\hat{A}$ in the sequel)



$q^{-1}(\{a\} \times X) =: X_a$ called Cameral curve

$A^\diamond \subseteq A$ those $a \in A$ w/ Cameral curve smooth & connected.

FACT: $\tilde{p}: \tilde{A}^\diamond \longrightarrow A^\diamond$ smooth of relative dim = 1

$\Rightarrow \tilde{p}^! \cong \tilde{p}^*[2]$. and for $\mathcal{F}_1, \mathcal{F}_2 \in D_c^b(A^\diamond, \mathbb{Q})$

$$\text{Hom}_{\tilde{A}^\diamond}(\tilde{p}^* \mathcal{F}_1, \tilde{p}^* \mathcal{F}_2) \cong \text{Hom}_{\tilde{A}^\diamond}(\tilde{p}^! \mathcal{F}_1, \tilde{p}^! \mathcal{F}_2)$$

$$\cong \text{Hom}_{A^\diamond}(\tilde{p}_! \tilde{p}^! \mathbb{Q} \otimes \mathcal{F}_1, \mathcal{F}_2) \cong \text{Hom}_{A^\diamond}(\tilde{p}_! \tilde{p}^! \mathbb{Q}_A, \underline{\text{Hom}}(\mathcal{F}_1, \mathcal{F}_2))$$

$$\text{i.e. } \phi: \tilde{p}^* \mathcal{F}_1 \longrightarrow \tilde{p}^* \mathcal{F}_2 \iff \phi^\sharp: \tilde{p}_! \tilde{p}^! \mathbb{Q}_A \longrightarrow \underline{\text{Hom}}(\mathcal{F}_1, \mathcal{F}_2).$$

$$c_1(L(\xi)): \underline{R p_{2*}}(R \hat{h}_* \mathbb{Q}_{\hat{M}^{\text{par}}}) \longrightarrow R p_{2*}(R \hat{h}_* \mathbb{Q}_{\hat{M}^{\text{par}}}[2])$$

$$= R \tilde{p}_* \circ R \tilde{f}_*$$

$$\text{And } R \tilde{f}_* \mathbb{Q}_{\hat{M}^{\text{par}}} = \tilde{p}^* R \hat{h}_* \mathbb{Q}_{\hat{M}}$$

$$\text{thus } c_1(L(\xi)): \tilde{p}^* R \hat{h}_* \mathbb{Q}_{\hat{M}} \longrightarrow \tilde{p}^* R \hat{h}_* \mathbb{Q}_{\hat{M}}[2]$$

$$\iff c_1(L(\xi))^\sharp: \tilde{p}_! \tilde{p}^! \mathbb{Q}_A \longrightarrow \underline{\text{Hom}}(R \hat{h}_* \mathbb{Q}_{\hat{M}}, R \hat{h}_* \mathbb{Q}_{\hat{M}}[2])$$